

Shell-Coded Casimir Boundary Transduction: Physical Readout Architectures for the White Sphere–Cylinder Geometry

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Abstract

White et al. introduced a custom Casimir geometry in which a micron-scale sphere inside a conducting cylinder produced a shell-like worldline-numerics vacuum response. A role-resolved follow-on audit separated that result into morphology, calibrated scale, readout participation, and ordinary material/electromagnetic backgrounds. The durable outcome is a shell-forming finite boundary object: the annular sphere–wall region remains the physical feature to interrogate, while direct central timing is assigned to an amplitude-bounded reference role and material response becomes calibration.

This paper develops the resulting physical continuation as a boundary-transduction problem. The experimental target is a geometry-locked Casimir response associated with the annular shell role, measured through ordinary precision apparatus channels with explicit material and electromagnetic controls. The central object is a shell-coded boundary response: a finite-geometry renormalized-stress organization whose physical evidence appears as force, pressure, resonant frequency, phase, or superconducting microwave response. The metric comparison supplies scale context; the apparatus claim is a measurable boundary response.

Three readout architectures are selected by the branch triage. A paired differential-pressure cell gives the direct stress/force channel and places patch-potential, roughness, membrane, thermal, and vibration control at the center of interpretation. A high-Q electromagnetic cavity converts boundary response into a frequency, phase, or mode-overlap observable. A superconducting resonator provides a cryogenic microwave path in which shell response can be separated from kinetic inductance, surface loss, vortex loading, quasiparticles, and material controls. Material impedance and capacitance measurements form the calibration channel that maps ordinary apparatus response.

The resulting program is precision finite-geometry Casimir metrology. A shell-coherent finding would identify a repeatable boundary-conditioned vacuum response in a finite shell-forming geometry and would supply a calibrated transducer for that response. Background-selected and sensitivity-bounded outcomes also carry value: they map how real machines turn geometry, materials, patches, resonator loss, and cryogenic conditions into measurable signals. The White sphere–cylinder system thereby becomes a benchmark for reading engineered Casimir boundaries through practical instruments.

1. INTRODUCTION

Casimir experiments translate boundary-conditioned quantum fields into measurable apparatus behavior. Conductors, dielectrics, superconductors, and structured finite geometries alter the electromagnetic mode spectrum. The renormalized stress associated with that altered spectrum appears in mechanical force, pressure, force gradients, resonant frequency shifts, phase shifts, quality-factor changes, and

temperature-dependent superconducting responses. Contemporary Casimir metrology is therefore an instrumentation discipline as much as a field-theory calculation: it depends on surface potential control, separation metrology, material characterization, roughness models, finite-conductivity corrections, resonator stability, and environmental rejection [4, 9, 5, 7].

White, Vera, Han, Bruccoleri, and MacArthur used worldline numerics to study custom Casimir cavities and reported a sphere-in-cylinder case whose negative energy-density morphology resembles the shell placement drawn in an Alcubierre-style source section [1, 2]. The compact case uses a $1\text{ }\mu\text{m}$ diameter sphere centered in a nominal $4\text{ }\mu\text{m}$ diameter cylinder. The geometry separates the stress-shaping body, inner wall, annular gap, material boundary, exterior control region, and possible readout channel into distinct experimental roles.

This paper treats the White sphere-cylinder system as a finite Casimir boundary object. The question is how the shell-coded boundary response can be read by a real instrument. The required architecture is a physical measurement with a geometry-indexed response vector. A viable readout must identify a transduction channel, measure material and electromagnetic backgrounds, modulate geometry, fit a shell template against nuisance templates, and give a physical interpretation for the recovered coefficient. Branch triage sets the route: amplitude-bounded central timing becomes a reference channel, material response becomes calibration, and pressure, cavity, and superconducting readouts become the physical continuation.

The readout program has three apparatus families. Differential pressure reads boundary stress directly through paired force cells. High-Q electromagnetic cavities read boundary response through resonant mode perturbation. Superconducting resonators read cryogenic boundary response through microwave frequency, phase, loss, and nonlinear mechanical channels. These branches are supported by existing experimental machinery: MEMS/NEMS Casimir force sensors [4, 7], high-Q nanomembrane force probes with Kelvin-probe surface-potential control [5, 6], long-range differential force measurements with explicit patch and roughness accounting [9], and superconducting Casimir or microwave-resonator platforms [8, 10, 11].

2. THE PHYSICAL CLAIM

The shell-coded response is a statement about renormalized stress-energy in a finite boundary value problem. In QFT, local energy density is represented by the renormalized expectation value of the stress-energy tensor. Boundary conditions can make that local value lower than the reference vacuum in some regions while averaged constraints keep the physical system bounded. Ford’s review describes this operational status of negative energy densities and the quantum inequality restrictions that accompany them [3].

Casimir apparatus makes this concept measurable. The negative sign in the ideal parallel-plate energy density expresses a boundary-conditioned stress relative to the reference state. The physical content is observed through force, pressure, energy, and mode behavior. Finite geometries add spatial structure: a boundary can concentrate the response in a specific region and the apparatus can couple to that region with varying strength.

The White sphere-cylinder geometry contributes a candidate spatial code. Its annular sphere-wall region acts as the shell role. The physical readout program asks whether that role can be coupled to a pressure cell, a cavity mode, or a superconducting resonator with enough strength and template separation for measurement against ordinary material and electromagnetic controls.

3. RELATION TO WHITE ET AL.

White et al. supplied a geometry, a worldline calculation, and a provocative metric comparison [1]. The present analysis separates the pieces needed for experimental use:

1. Geometry: a small, finite, role-structured sphere–cylinder boundary.
2. Morphology: a shell-like organization of the vacuum-boundary response.
3. Scale: the calibrated stress level associated with that response.
4. Transduction: the physical apparatus channel that converts the response into measured data.

The first two pieces specify the controlled geometry: the shell band is the region to modulate, and the body-wall spacing is the coordinate that should move the shell template. The calibrated scale piece assigns direct central timing and direct metric readout to an amplitude-bounded reference role. Material and impedance observables become calibration channels because they measure ordinary capacitance, contact, conductivity, and loss responses that share geometry controls with the shell. Differential pressure carries the direct stress measurement, with interpretation governed by patch-potential, roughness, membrane, and common-mode rejection. High-Q cavity and superconducting resonator readouts carry the resonant route: they convert boundary perturbation into frequency, phase, mode-splitting, or microwave response while retaining nuisance templates in the same fit.

4. READOUT MODEL

Let q denote a controlled geometry or apparatus setting: gap, sphere radius or surrogate radius, cylinder radius, wall thickness, lateral offset, cell length, mode overlap, readout radius, temperature, magnetic history, or drive power. The measured observable $y(q)$ is modeled as

$$y(q) = a_{\text{shell}}T_{\text{shell}}(q) + \sum_i a_i T_i(q) + \varepsilon(q), \quad (1)$$

where T_{shell} is the shell-coded response template and the T_i are measured background templates. The background family includes patch potential, capacitance, roughness, membrane drift, temperature, vibration, finite conductivity, waveguide dispersion, mode mixing, kinetic inductance, surface loss, vortex response, quasiparticles, and material controls.

Equation 1 defines the evidential standard. The result is a shell coefficient with uncertainty, a set of nuisance coefficients with uncertainties, and a false-positive check from material-only and electromagnetic-only control schedules. A single absolute shift has limited meaning in this apparatus class. A geometry-locked response vector can separate shell, material, and readout behavior.

The design-screening ladder associated with this paper used a recovery gate of $\text{SBR} = 1.2$ and an electromagnetic-only false-positive target near $\text{FP} < 0.05$. Table 2 summarizes the portable design implications.

5. COMMON APPARATUS ARCHITECTURE

All three readout branches require the same experimental skeleton. The apparatus contains a shell-sensitive cell, a matched dummy cell, a material-control channel, a geometry actuator, surface metrology, and a readout chain with environmental monitors. The shell cell implements the sphere–cylinder geometry or a manufacturable surrogate preserving the same shell-overlap coordinates. The dummy cell preserves material, wiring, thermal mass, and readout coupling while suppressing shell overlap. The material channel measures impedance, capacitance, contact, and loss behavior across the same schedule.

The geometry schedule is part of the measurement. Gap, radius, offset, wall thickness, length, and readout overlap should move the shell template differently from patch, capacitance, material, and drift templates. Surface measurements should accompany those settings. Garcia-Sanchez et al. show how Kelvin-probe surface-potential mapping can be integrated into high-Q nanomembrane Casimir measurements [5]. Bimonte et al. demonstrate a long-separation force measurement with explicit treatment of patch potentials, roughness, edge effects, uncertainty, and theory comparison [9].

Table 1: Physical readout families for the shell-coded boundary response.

Family	Observable	Physical transduction	Main control role
Differential pressure	Paired pressure or force contrast	Boundary stress in the shell-gap geometry	Patch, roughness, membrane, thermal, and vibration rejection
High-Q cavity	$\delta f/f$, phase, Q , or mode splitting	Electromagnetic eigenmode perturbation from shell-sensitive boundary overlap	Loss, thermal, waveguide, and mode-mixing separation
Superconducting resonator	Microwave $\delta f/f$, phase, Q , nonlinear drum response, or T_c -coupled response	Cryogenic boundary response in a superconducting microwave apparatus	Surface loss, kinetic inductance, vortex, quasiparticle, and material separation
Material impedance	Capacitance, impedance, contact, or loss response	Ordinary material and electromagnetic apparatus behavior	Calibration map for backgrounds

Table 2: Screening-level design constraints for the physical branches.

Branch	Representative viable cases	Predicted SBR range	Design implication
High-Q cavity	nominal clean through optimistic clean/matched/lossy	2.22–19.34	Strongest design row; mode overlap and loss templates set interpretation
Superconducting resonator	nominal balanced/clean through optimistic clean/balanced/lossy	3.59–17.76	Strong cryogenic path; surface-loss and quasiparticle controls are central
Differential pressure	optimistic balanced and optimistic quiet	2.02–5.31	Direct force path; patch and common-mode controls set feasibility
Material impedance	calibration rows	background map	Measures ordinary apparatus response for the same geometry

The instrument must first demonstrate ordinary Casimir sensitivity in a known geometry. This calibration step establishes separation metrology, electrostatic calibration, roughness priors, and noise floors. The shell geometry then becomes a finite-boundary test object whose output is judged by template selection and control behavior.

6. DIFFERENTIAL-PRESSURE ARCHITECTURE

The differential-pressure architecture reads boundary stress directly. Two matched cells share environment and readout while their geometry settings produce different shell-gap responses. The primary observable is

$$\Delta P(q) = P_{\text{shell}}(q) - P_{\text{control}}(q), \quad (2)$$

with the fit model of Eq. 1. For ideal parallel plates,

$$P_C(a) = -\frac{\pi^2 \hbar c}{240a^4}, \quad (3)$$

which gives about 1.3×10^{-3} Pa at $a = 1 \mu\text{m}$. The sphere–cylinder cell is finite and curved, so Eq. 3 serves as a scale reference for the apparatus model. The screening rows place the strongest pressure cases in the same metrological neighborhood, with shell pressures around 10^{-4} – 10^{-3} Pa.

The machinery is close to established Casimir force metrology. Decca, Aksyuk, and Lopez review micro- and nano-electromechanical systems for Casimir measurements [4]. Stange et al. show a practical commercial MEMS force platform with piconewton-scale force resolution and electrostatic calibration [7]. Garcia-Sanchez et al. demonstrate high-Q nanomembrane force-gradient readout with in situ surface-potential control [5, 6]. These examples define the required toolchain: calibrated actuation, separation sensing, patch mapping, roughness measurement, and common-mode rejection.

A pressure result would identify the shell-coded boundary response as a finite-geometry force or stress contrast. The immediate utility is precision microforce calibration, patch-potential diagnostics, roughness validation, and MEMS/NEMS pressure metrology at micron gaps.

7. HIGH-Q CAVITY ARCHITECTURE

The high-Q cavity architecture uses resonance as the transducer. A cavity mode with large field overlap in the shell-sensitive region converts a small boundary perturbation into a measurable frequency, phase, Q , or mode-splitting signal. In first-order perturbation language,

$$\frac{\delta f}{f} \sim -\frac{1}{2} \frac{\int_{\Omega} (\delta\epsilon |E|^2 - \delta\mu |H|^2) dV}{\int_{\Omega} (\epsilon |E|^2 + \mu |H|^2) dV} + \Delta_{\text{boundary}}, \quad (4)$$

where Δ_{boundary} denotes the shell-sensitive finite boundary contribution modeled by the apparatus eigenmode solver. The equation states the design requirement: maximize known shell overlap and measure the ordinary material terms in the same basis.

The screening ladder ranks this branch highly. The strongest cavity row gives $\text{SBR} = 19.34$, and the nominal clean row remains above the recovery gate. Those values require measured templates for conductor loss, thermal expansion, waveguide dispersion, port coupling, and mode mixing. A strong cavity design therefore includes a shell-overlap mode, a low-overlap or null mode, a dummy cavity, independent thermal metrology, deliberate port-coupling sweeps, and a validated eigenmode model.

The positive interpretation is electromagnetic mode transduction of the shell-coded boundary response. Such a result would support high-Q boundary-sensitive sensors, finite-geometry cavity calibration, and experimental validation of Casimir calculations in resonant structures.

8. SUPERCONDUCTING-RESONATOR ARCHITECTURE

The superconducting architecture uses a cryogenic microwave resonator, drum, or transition-sensitive cavity to read the shell-coded response. The observable can be frequency shift, phase shift, Q change, nonlinear mechanical response, or a superconducting transition response. The core fit is

$$\left(\frac{\delta f}{f}\right)_{\text{sc}} = a_{\text{shell}}T_{\text{shell}} + a_{\text{ki}}T_{\text{ki}} + a_{\text{surf}}T_{\text{surf}} + a_{\text{vort}}T_{\text{vort}} + a_{\text{qp}}T_{\text{qp}} + a_{\text{mat}}T_{\text{mat}} + \varepsilon, \quad (5)$$

with kinetic inductance, surface loss, vortex loading, quasiparticles, and material response treated as measured templates.

This branch has unusually strong source support. Perez et al. designed a NEMS system to probe Casimir energy corrections to superconducting condensation energy by measuring a superconducting film in a controlled cavity [8]. de Jong et al. report a superconducting Casimir force measurement through nonlinear dynamics of a superconducting drum resonator in a microwave optomechanical system [11]. Gurevich reviews the microwave-loss mechanisms that govern superconducting resonators, including quasiparticles, residual surface resistance, dielectric loss, trapped vortices, kinetic inductance effects, and high-field limits [10]. Those mechanisms are the exact nuisance family in Eq. 5.

The screening ladder gives superconducting rows from $\text{SBR} = 3.59$ to $\text{SBR} = 17.76$ across the viable nominal and optimistic cases. The apparatus should include shell-coupled and material-only resonators on the same cooldown path. Temperature sweeps across and below T_c , magnetic-history tests, vortex-loading controls, drive-power sweeps, quasiparticle checks, and geometry modulation all become evidence-bearing coordinates.

A superconducting result would identify a cryogenic boundary response separated from microwave-loss physics. Its practical value is superconducting surface-loss diagnostics, cryogenic vacuum-gap metrology, microwave boundary-sensing, and experimental tests of superconducting electrodynamics in finite Casimir geometries.

9. CONTROL MATRIX AND OUTCOME CLASSES

The control channels are measured observables. They define the interpretation of the shell coefficient. Table 3 summarizes the control standard for each physical branch.

Table 3: Evidence-bearing controls for physical readout branches.

Branch	Required controls	Shell-coherent data signature
Differential pressure	Matched dummy cell, opposite gap schedule, Kelvin-probe map, roughness prior, membrane drift model, thermal and vibration monitors	Pressure vector selects the shell schedule after patch, roughness, and membrane templates are included
High-Q cavity	Eigenmode model, shell-overlap and low-overlap modes, dummy cavity, thermal expansion monitor, loss and port-coupling sweeps	Frequency or phase vector selects shell overlap over loss, thermal, waveguide, and mode-mixing templates
Superconducting resonator	Material-only resonator, temperature sweep, drive-power sweep, magnetic-history control, quasiparticle checks, surface-loss model	Microwave response selects shell schedule after kinetic-inductance, loss, vortex, quasiparticle, and material templates are included

Each outcome supplies a different experimental product. Shell coherence opens a dedicated apparatus design paper. Material or resonator selection improves the background map. A sensitivity bound sets a quantitative design target for the next transducer.

Table 4: Outcome classes for shell-geometry measurements.

Outcome	Data signature	Physical meaning
Shell coherent	Shell coefficient exceeds recovery gate and controls satisfy the false-positive target	Finite White-style boundary geometry produces a measurable shell-coded Casimir response in that transducer
Material selected	Patch, capacitance, impedance, contact, or loss template has the stable coefficient	The apparatus maps ordinary material/electromagnetic response of the same geometry
Resonator selected	Cavity or superconducting response follows mode mixing, conductor loss, surface loss, vortex, or quasiparticle template	Resonator physics dominates the readout channel under the tested settings
Sensitivity bounded	Shell and nuisance templates remain below recovery gate after calibration	The apparatus sets an upper bound on shell-coded transduction for that branch

10. PHYSICAL IMPLICATIONS

A pressure result would identify a direct finite-geometry stress response. It would connect the shell-coded morphology to a mechanical observable and would test how patch potentials, roughness, edge structure, and common-mode rejection interact with finite Casimir geometry.

A high-Q cavity result would identify resonant electromagnetic transduction. The shell response would appear as an eigenmode perturbation with measurable geometry dependence. In the design screen, this branch supplies the largest predicted signal-to-background row, so its apparatus priority comes from mode-overlap leverage and measurable loss templates.

A superconducting resonator result would identify a cryogenic boundary response separated from superconducting loss channels. This branch is the branch where superconducting Casimir force measurements, condensation-energy questions, microwave resonator loss, and quantum-limited sensing can be tested in a shared cryogenic microwave language.

Across all branches, the physics is finite-boundary QFT metrology. The measured quantity is the consequence of boundary-conditioned renormalized stress as read through a calibrated apparatus channel. The metric comparison remains a scale reference and a motivation for the shell geometry.

11. OUTCOME-DEPENDENT PRACTICAL UTILITY

The practical utility of the program depends on which coefficient the White-derived geometry schedule selects. The schedule moves shell gap, material wall, central region, exterior control, and readout overlap as separate coordinates. A finding in this setting carries more information than a raw force or frequency shift because it identifies which coordinate couples to which physical transducer.

A shell-coherent pressure result would create a finite-geometry force and pressure calibration object. The apparatus would provide a measured micron-gap stress contrast tied to the annular shell role, with patch potential, roughness, membrane drift, edge structure, and common-mode subtraction recorded in the same dataset. The utility would be strongest for MEMS/NEMS force standards, surface-potential diagnostics, and validation of curved and finite-boundary Casimir calculations. The same data would quantify how much residual electrostatic structure remains after surface-potential mapping and paired-cell subtraction.

A shell-coherent high-Q cavity result would create a resonant boundary-transduction standard. The measurement product would be a geometry-indexed frequency, phase, or mode-splitting response

whose shell coefficient is separated from conductor loss, thermal expansion, waveguide dispersion, and mode mixing. That finding would support compact sensors for micron-scale gap changes, boundary motion, surface-condition changes, and finite-geometry electromagnetic perturbations. It would also give numerical Casimir methods a new resonant observable to predict, beyond direct force.

A shell-coherent superconducting resonator result would connect the White shell geometry to microwave resonator physics while separating the shell coefficient from kinetic inductance, surface loss, vortex loading, quasiparticles, and material controls. The practical outputs would include superconducting surface-loss diagnostics, vortex and quasiparticle hygiene tests, cryogenic gap metrology, and a controlled platform for superconducting Casimir force or condensation-energy studies. This branch would also help determine which parts of a superconducting microwave response are geometric boundary effects and which parts are material-loss physics.

A material-, patch-, or loss-selected result would produce an artifact atlas for a hard finite Casimir geometry: how patch potentials, capacitance, contact loading, finite conductivity, thermal drift, waveguide coupling, surface loss, and superconducting quasiparticles mimic or obscure shell-coded structure. That outcome would improve future false-positive rejection and would tell apparatus builders which background channel dominates each readout family.

A sensitivity-bounded result would turn the White-derived program into an engineering bound. The measured upper limits would specify how much more area, quality factor, common-mode rejection, gap stability, surface control, or cryogenic loss suppression is required for each branch. The same shell geometry then stresses force, resonant, and superconducting readouts in parallel.

The integrated utility is conditional and cumulative. A positive shell result gives a new finite-geometry Casimir transducer. A background selection gives a calibrated map of ordinary apparatus physics in a difficult geometry. A sensitivity bound gives a quantitative design target. Together, those outcomes define a practical benchmark for how engineered Casimir boundaries can be read through real machines.

12. REPRODUCIBILITY AND SOURCE BASIS

The citable scientific basis for this manuscript is the public literature in the reference list. The references provide the canonical DOI, arXiv, NIST, and journal locations for the source material.

The numerical screening values quoted in Tables 2 and 3 come from a finite readout-ladder calculation using shell-template recovery, electromagnetic-only false-positive checks, and physical backend cases for pressure, cavity, superconducting, material, and central-reference channels. The associated supplementary material consists of compact machine-readable screening summaries: candidate definitions, backend physical cases, gate summaries, and run-level summary metadata. Archiving the manuscript source and PDF with those summaries keeps the quoted design-screening values traceable.

13. CONCLUSION

The White sphere–cylinder geometry now has an experimental role map. The sphere and inner cylinder wall set the finite boundary condition. The annular gap carries the shell template. The material wall, exterior region, and central channel supply control and reference roles. The prior audit maps those roles onto apparatus choices: shell morphology motivates modulation of the sphere–wall gap, calibrated scale makes central timing an amplitude-bounded reference, material and impedance response become calibration, and the remaining physical question is transduction.

Pressure, cavity, and superconducting readouts interrogate different parts of that transduction problem. A pressure apparatus asks whether paired cells can recover a shell-indexed stress contrast after patch maps, roughness priors, membrane drift, and common-mode subtraction are included. A high-Q cavity asks whether the same shell coordinate moves an electromagnetic eigenmode in a way

that separates from conductor loss, thermal expansion, waveguide coupling, and mode mixing. A superconducting resonator asks whether a cryogenic microwave device can carry the shell coefficient after kinetic inductance, surface loss, vortex loading, quasiparticles, and material controls are measured in the same fit.

The outcome classes are concrete. Shell coherence would create a finite Casimir boundary transducer tied to a specific physical channel. Background selection would produce a measured atlas of the ordinary material, patch, loss, and resonator effects that share the White geometry's controls. Sensitivity bounds would specify the next apparatus requirements: area, quality factor, gap stability, surface control, common-mode rejection, and cryogenic loss suppression. These outcomes are cumulative because they use the same shell geometry and the same template schedule.

The program therefore converts the original visual metric comparison into a portable metrology problem. The sphere–cylinder device becomes a benchmark for reading engineered Casimir boundaries through real machines. The next physical result should identify where the shell coefficient lands: in direct pressure, in electromagnetic mode shift, in superconducting microwave response, or in a calibrated background channel that fixes the next design bound. That answer would sharpen how engineered vacuum-boundary structure becomes measurable force, pressure, frequency, phase, or cryogenic resonator response.

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